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Original Report

# Quality standards in radiation medicine

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**Abstract**

**Purpose:** Identifying and conducting "best practice" medicine is arguably the ubiquitous goal of practitioners. However, to distill the many available quality standards, guidelines, recommendations, and indicators down to a best practice set requires a logical schema to group standards addressing similar quality issues and, from manageable lists of related standards, to extract the essential dimensions of quality. The purpose of this study was to explore a method of collating publicly available quality standards, in this case in radiation therapy, using a 2-step decision tree approach with statistical analysis. Successful grouping into manageable lists, addressing related quality issues, informs the ongoing development of quality indicators that are one expression of "best practice."

**Methods and materials:** A comprehensive literature search was used to identify quality standards currently in use and publicly available. Using 2 decision trees, 5 evaluators assigned each standard to Donabedian's structure, process, or outcome and also to the target of the standard: patients, staff, equipment or clinical process, or organization for a total of 3 × 4 = 12 primary categories.

**Results:** A total of 454 radiation medicine program quality standards spread across 8 national and international documents was identified. Agreement between the 5 evaluators, using the free marginal kappa statistic, ranged from fair to almost perfect. In all but 2% of 5 × 454 evaluations were the evaluators able to assign a statement to categories in the decision trees suggesting that these trees are appropriate to the task. In only 3/454 was a majority (≥ 3/5) decision not reached on the assignment to structure, process, or outcome. Sixty-four percent of the standards were identified with structure, 26% with process and 10% with outcome.

**Conclusions:** Donabedian's model constitutes a reliable method of managing quality standards. The 2-step decision tree framework can be applied to inform the further development of national and international quality standards.



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## Introduction

Much of the attention of the radiation medicine community at large has, in recent years, been focused on the safety of the care we deliver.[1] It could be argued that our emphasis on rare but tragic events has overshadowed an equally significant problem, that of suboptimal quality leading to suboptimal outcomes.[2,3] Although we view the link between quality and outcome as axiomatic, firm evidence of the relationship is exceedingly scarce. It is only now that results are beginning to appear in the clinical trials literature of the connection between protocol noncompliance, ie, implied suboptimal quality, and patient outcomes.[3,4] If we accept that there is a potential quality problem in radiation medicine the question then becomes, "what to do about it to achieve best practice?" Several authors have presented thoughtful discussions of what is meant by quality in radiation medicine, what are the dimensions of quality, and how we might develop quality metrics.[5-7] Much of the detailed quality standard work to date has been disease specific[8] with the more generic approaches to evaluating quality in a radiation medicine program as a whole being embodied in accreditation standards and guidelines.[9] When applying best practice principles for guideline development, documents are usually informed by a formal consensus process and the evidence base.[10] However, the step of identifying and sorting what constitutes appropriate quality standards for redeveloped guidelines is less well explored.

In 2011, The Canadian Partnership for Quality Radiotherapy (www.cpqr.ca) was established with a broad mandate of leading and facilitating quality and safety initiatives in radiation medicine across Canada. The CPQR has already issued draft *Quality Assurance Guidance for Canadian Radiation Treatment Programs* and is well into the process of updating previously existing technical standards documents. The guiding principles in updating and maintaining these key documents are that they should be scientifically sound, evidence based where possible, unambiguous, and relevant. Acknowledging that much work in the area of radiation treatment quality is ongoing in other jurisdictions, it is clearly prudent for CPQR and other organizations interested in quality standards to access the expertise and efforts of others in the continuing development of standards, guidelines, and indicators. The challenge, however, is to organize the many and disparate standards emanating from a variety of national and international bodies within a logical structure that facilitates analysis and informs the ongoing development of quality documents.

The purpose of this study was to develop and validate a novel decision tree approach to the collation of publicly available radiation treatment quality standards, guidelines, recommendations, and indicators (denoted hereafter collectively as standards). Examining a large unsorted list of quality standards for duplicates, distilling the salient quality measures, and hence identifying gaps in the Canadian, or any other document, is a virtually impossible task without grouping them into some logical categories, which would then contain fewer standards and be much more amenable to gap and content analysis.

While the purpose of this manuscript is to describe our experience with this approach to the distillation of quality standards, the interested reader may access the extensive detailed collation through an electronic appendix (available as supplementary material online only at www.practicalradonc.org).

This study outlines a practical method to collate, catalogue, and analyze available quality standards for use as input to a formal assessment of expert opinion, through a modified Delphi process for example, for guideline redevelopment.

## Methods and materials

Relevant program quality documents were traced using Google Scholar and Discover. The criteria for selection were the following: (1) national or international; (2) contain statements or questions related to measuring quality in a broad sense in radiation medicine programs; and (3) written in English. Specifically technical quality control standards were not included in this analysis. In 8 publicly available documents, 454 quality standards or groups of standards were identified (Table 1). One of these documents was CPQR's draft *Quality Assurance Guidance for Canadian Radiation Treatment Programs.* As an illustration, Table 2 lists 5 typical standards that the evaluators were requested to assign according to the scheme described below.

Donabedian's framework[11] is widely used in health care quality assessment[6,12] and has been adopted here for the first step in categorizing the 454 statements or standards. Thus, the analysis required that each statement be sorted according to Donabedian's structure, process or outcome.[11] Had the sorting procedure been terminated at this point, the 3 resultant groups would still have been too large (~ 150 standards each) to easily digest. Hence a further sorting by the target group of the standard (ie, patients, staff, equipment or clinical process, or organization) was performed. This 2-step sorting process resulted in grouping of the 454 standards into $3 \times 4 = 12$ primary classifications (average size: 40 standards), which not only makes the identification of duplicate standards among the 8 source documents manageable but also facilitates the extraction of the key quality measures being addressed in the 8 documents.

Five evaluators, all authors of this paper, performed the 2-step categorization using a spreadsheet to record their assignments. The process followed by the 5 evaluators is elucidated in Fig 1A and B that show the decision trees used to assign the standards to the major and minor categories.

**Table 1** Source documents

| Data Source | Website | Document | Country/region |
|---|---|---|---|
| American College of Radiation Oncology (ACRO) | www.acro.org | Radiation Standards Medical Physics (external beam therapy) | United States |
| American College of Radiology (ACR) | www.acr.org | Practice Guideline For Radiation Oncology | United States |
| Canadian Partnership for Quality Radiation Therapy | www.partnershipagainstcancer.ca | Quality Assurance Guidance for Canadian Radiation Treatment Programs | Canada |
| European Commission Guideline on Clinical Audit (RT sections only) | http://ec.europa.eu/index_en.htm | European Guidelines on Clinical Audit for Medical Radiological Practices (Diagnostic Radiology, Nuclear Medicine and Radiotherapy) | Europe |
| IAEA (Quality Assurance Team for Radiation Oncology (QUATRO) | www.iaea.org | Comprehensive Clinical Audits of Diagnostic Radiology Practices: A Tool for Quality Improvement | International |
| National Cancer Review Programme Manual for Cancer Services (NCAT) | www.ncat.nhs.uk | Manual for Cancer Services 2008: Radiotherapy Measures (Version 2) | United Kingdom |
| National Patient Safety Agency (NHS) | www.hpa.org.uk | Toward safer radiotherapy: a self-assessment tool | United Kingdom |
| Royal Australian and New Zealand College of Radiologists (RANZCR) | www.ranzcr.edu.au | Tripartite Radiation Oncology Practice Standards | Australia, New Zealand |

In some of the 454 standards there were 1 or more substatements. To keep the project manageable only the main headings were used in the analysis. Thus, through this 2-step process each quality standard was assigned to 2 categories. Category 1 with 3 primary options, the major category, was based on Donabedian's structure, process and outcome.[11] The minor 4 primary option category, category 2, described the "target" of the quality statement and comprised patient, staff, equipment or clinical process, and organization. "Other" was added to both category 1 and category 2 to provide evaluators with an option if the proposed taxonomies did not appear comprehensive enough. This resulted in 4 × 5 = 20 classifications.

The spreadsheets, filled out independently by the 5 evaluators, were examined for interrater reliability using the Fleiss free marginal kappa.[13] Fleiss' kappa can be used when there are more than 2 raters and the object of the evaluation is "nominal categorical," such as the Donabedian

**Table 2** Examples of standards and assignments

| Standard | Category 1 (No. of evaluators) | Category 2 (No. of evaluators) | Final disposition |
|---|---|---|---|
| The aims of the department must be clearly defined and the infrastructure, resources and practice consistent with achieving and sustaining these aims | Structure (5) | Organization(5) | Structure/Organization |
| Informed consent for radiation treatment is obtained at the time that the decision to treat is finalized | Process (5) | Patient (5) | Process/Patient |
| Documented quality control activities that evaluate feasibility and suitability of the proposed treatment plan | Process (5) | Organization (1)<br>Equip/Process (2)<br>Patient (2) | Process/Unassigned |
| Ethics approval of all clinical trials from a committee in accordance with …… guidelines | Structure (3)<br>Process (1)<br>Outcome (1) | Organization (2)<br>Equip/Process (2)<br>Patient (1) | Structure/ Unassigned |
| Documentation that the facility has successfully participated in an external dosimetric intercomparison conducted with a nonaffiliated organizationally separate service within the last 2 years and which has been reviewed and actioned as appropriate | Outcome (5) | Organization (4)<br>Equip/Process (1) | Outcome/Organization |

classifications.[14] Two common types of Fleiss' kappa can be used: free or fixed marginal. Free marginal kappa is appropriate in this instance as raters did not have to distribute their classifications in a certain way (eg, 25% into structure). Although it is a conservative measure of rater concordance, this statistic is useful to demonstrate reliability of the 2-step decision tree approach to obtain classification agreement for the 454 standards.

The analysis was performed for categories 1 (Donabedian) and 2 (target) separately and together. This first iteration is referred to as exercise 1. As described below, agreement between the evaluators ranged from fair to moderate based on

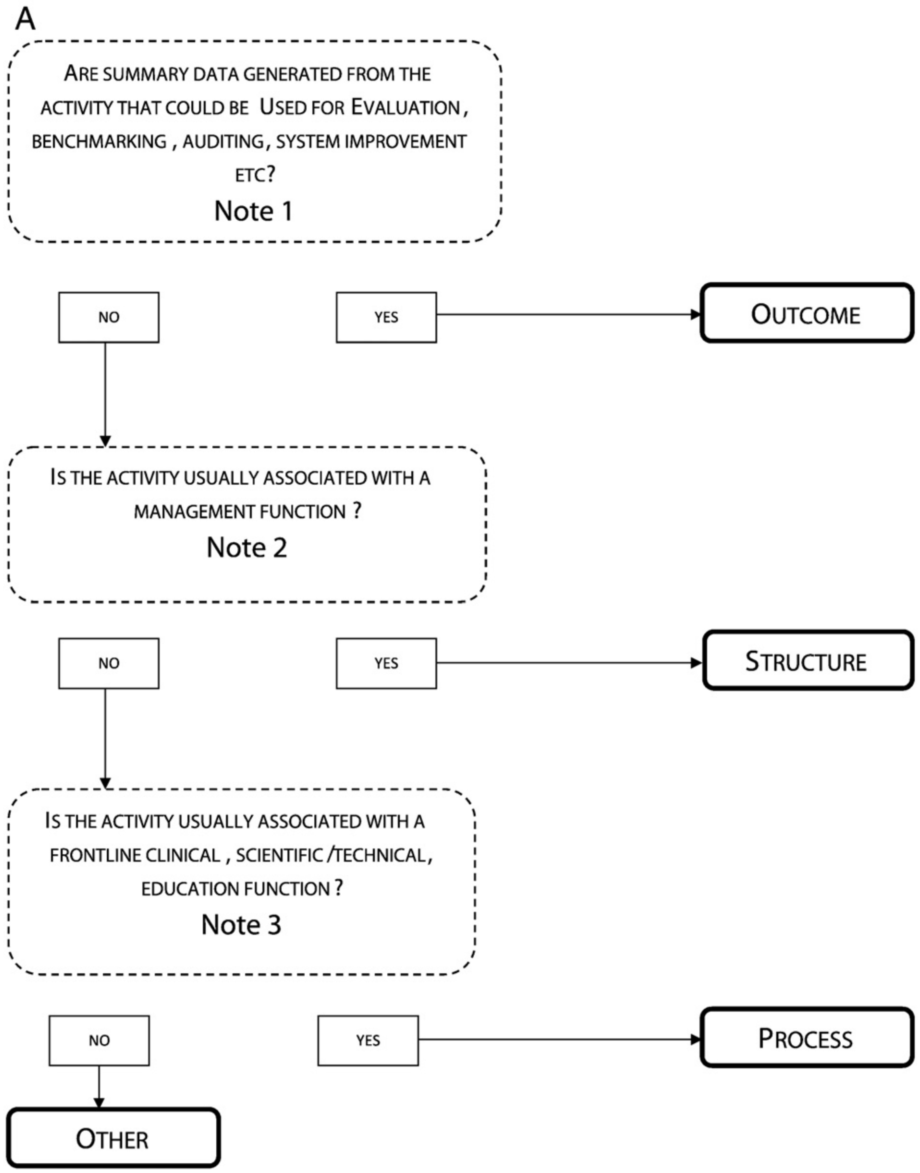


**Figure 1** (A) Category 1 (Donabedian) decision tree. (Note 1) A standard that includes the words "audit", "evaluation" or "review" may imply that summary data are generated which are or could be used as an intermediate outcome metric; records of training and attendance at rounds, etc imply the existence of intermediate outcome metric; comparison with national guidelines or standards of practice implies the existence of intermediate outcome metric. (Note 2) Policies or procedures related to resources and resource allocation are structural; activities that shape the culture of the organization are structural; activities described in the policy or procedure that would be performed as a management function (ie, wearing a management hat) are structural; activities performed by a department, program, or committee are structural; activities that are volume independent are likely to be structural (eg, operating a radiation safety program is necessary irrespective of the numbers of patients, staff or machines); (Note 3) If the policy or procedure addressed in the standard is for use during a front line function, irrespective of who developed it, it is process; activities that are volume dependent are likely to be process (eg, checking patients' identification or machine quality control) will be carried out more often if there are more patients or more machines. (B) Category 2 (target) decision tree.

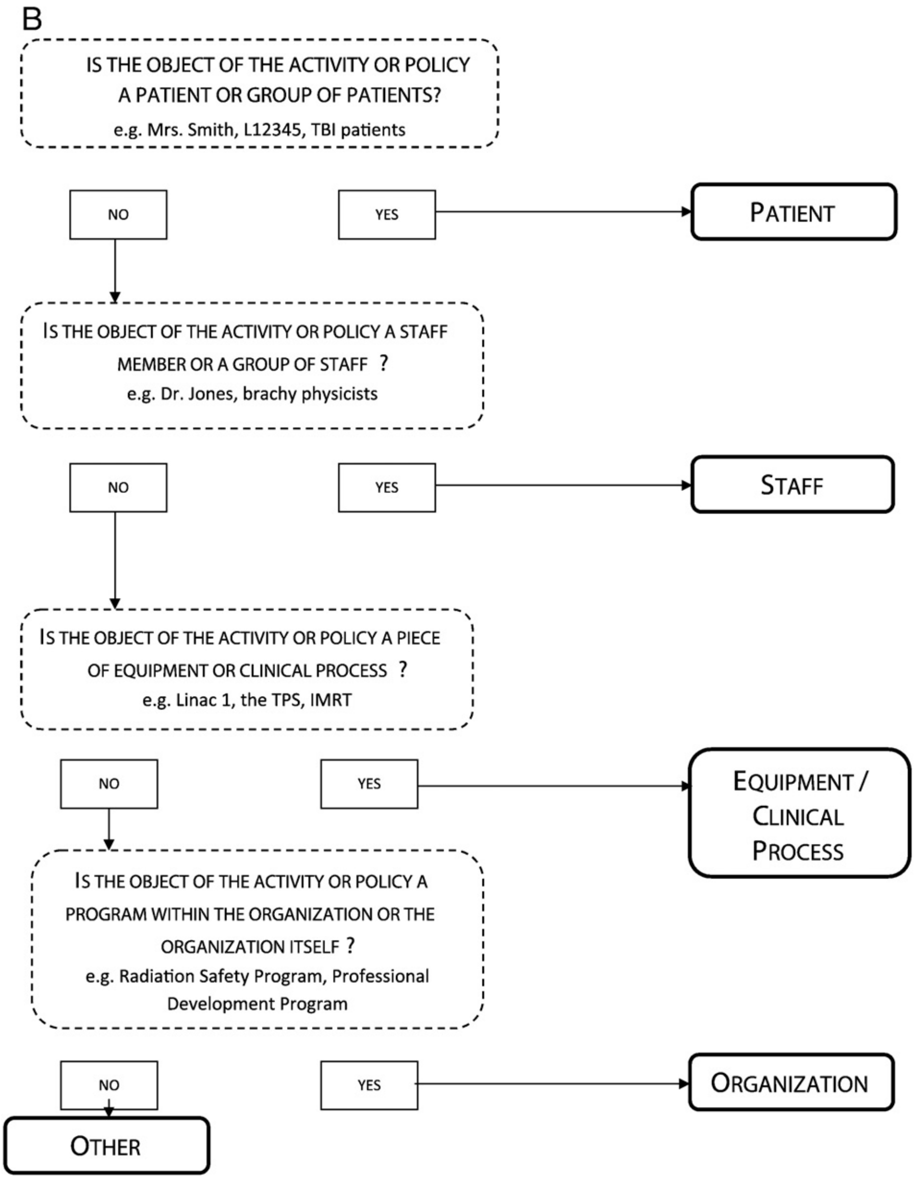


**Figure 1** (*continued*)

kappa criteria.[15] However, evaluators reported requiring 6 to 8 hours to complete the assignment of all 454 quality standards and all reported a level of fatigue. Those indicators that did not achieve 5/5 (ie, complete) agreement in category 1 (Donabedian) were isolated and submitted again to the independent evaluators with the hope of minimizing the influence of fatigue and hence possibly obtaining greater consistency. This second iteration on category 1 (Donabedian) only is referred to below as exercise 2.

## Results

Discussing first exercise 1, in which both categories 1 (Donabedian) and 2 (target) were assigned, agreement between the evaluators was either fair (κ from 0.21 to 0.40) or moderate (κ from 0.41 to 0.60) (Table 3). Viewing the results another way, 3 or more reviewers assigned the same categories 1 and 2 to 67% of the 454 quality standards, which is noteworthy considering the variability in the grammatical structures adopted by the 8 documents. Looking at agreement for category 1 assignments only, it is not surprising that agreement improved. Agreement on the interpretation of 1 document was substantial (κ from 0.61 to 0.8), 6 were rated as moderate and the eighth as fair. Three or more evaluators agreed on the assignment within category 1 (Donabedian) for 93% of the quality statements.

Exercise 2 involved having the 5 evaluators reexamine all standards with less than 5/5 agreement (n = 290) in exercise 1. To mitigate the evaluator fatigue that may reduce consistency, this was limited to category 1. Combining the 290 reevaluations with the 164 statements

**Table 3** Free marginal kappa for exercises 1 and 2

| Source | Exercise 1: | | Exercise 2: | |
|---|---|---|---|---|
| | Categories 1 & 2 (κ) | Category 1 only (κ) | Category 1; revised & 2 original (κ) | Category 1 only; revised (κ) |
| All | 0.38 (Fair) | 0.54 (Moderate) | 0.45 (Moderate) | 0.71 (Substantial) |
| American College of Radiation Oncology (ACRO) | 0.48 (Moderate) | 0.48 (Moderate) | 0.58 (Moderate) | 0.65 (Substantial) |
| American College of Radiology (ACR) | 0.36 (Fair) | 0.54 (Moderate) | 0.41 (Moderate) | 0.67 (Substantial) |
| Canadian Partnership for Quality Radiotherapy | 0.45 (Moderate) | 0.59 (Moderate) | 0.54 (Moderate) | 0.81 (Almost perfect) |
| European Commission Guideline on Clinical Audit (RT sections only) | 0.35 (Fair) | 0.57 (Moderate) | 0.42 (Moderate) | 0.70 (Substantial) |
| IAEA (Quality Assurance Team for Radiation Oncology (QUATRO) | 0.41 (Moderate) | 0.48 (Moderate) | 0.46 (Moderate) | 0.61 (Substantial) |
| National Cancer Review Programme Manual for Cancer Services (NCAT) | 0.34 (Fair) | 0.59 (Moderate) | 0.40 (Moderate) | 0.77 (Substantial) |
| National Patient Safety Agency (NHS) | 0.43 (Moderate) | 0.61 (Substantial) | 0.50 (Moderate) | 0.75 (Substantial) |
| Royal Australian and New Zealand College of Radiologists (RANZCR) | 0.28 (Fair) | 0.35 (Fair) | 0.42 (Moderate) | 0.66 (Substantial) |

that had 5/5 agreement in exercise 1 yielded a much improved free marginal kappa of 0.71, which falls into the range of substantial agreement (κ from 0.61 to 0.8). Working on the underlying assumption that the target of the standard would not change if its Donabedian classification did (ie, that the 2 categories are independent of one another), category 2 classifications from exercise 1 were added to the reevaluated category 1 classifications. This resulted in the Fleiss' kappa rising to 0.45 (moderate agreement) as shown in Table 3.

The division of the 454 standards across categories 1 and 2 is shown in Table 4. Statements were assigned to the categories selected by 3 or more of the evaluators. Where a simple majority (ie, 3 or more evaluators) failed to agree on an assignment the result has been recorded as unassigned in Table 4.

Table 2 gives examples of standards for which there was a high degree of unanimity and standards for which interpretation was very variable.

In the 5 × 454 assignments made during exercise 2, most were classified as structure for category 1 (64%), followed by process (26%), and then outcome (10%), as shown in Table 4. It is interesting, but perhaps not surprising, that the numbers of standards with patients or staff as their foci are overshadowed by those addressing equipment or clinical process and organizational issues. With increasing emphasis on evidence-based medicine and patient-centered care, together with the identification of inadequate training as a significant cause of radiation therapy misadministration, the distribution of standards among category 2 options could change as the quality agenda moves forward.

The final spreadsheet in which all 454 standards are assigned to categories 1 and 2 is available from the electronic archive.

## Discussion

The primary purpose of this study was to test the feasibility of using a 2-step decision tree to classify a large number of standards into manageable lists that could then be examined for duplicates and the essential elements could be distilled. The fact that so few evaluations (2%) were not answered or assigned to the category of "other" suggests that the decision trees presented in Fig 1A and B include a comprehensive range of categories. In only 3

**Table 4** Division of assignments between categories 1 and 2 following exercise 2

| Category 2 assignment | Equipment/clinical process | Organization | Patient | Staff | Unassigned | Total | % Total |
|---|---|---|---|---|---|---|---|
| Category 1 assignment | | | | | | | |
| Structure | 106 | 79 | 15 | 68 | 22 | 290 | 63.9 |
| Process | 69 | 3 | 28 | 6 | 10 | 116 | 25.6 |
| Outcome | 12 | 8 | 8 | 5 | 12 | 45 | 9.9 |
| Unassigned | 1 | 1 | 0 | 0 | 1 | 3 | 0.7 |
| Grand total | 188 | 91 | 51 | 79 | 45 | 454 | 100.0 |
| % total | 41.4 | 20.0 | 11.2 | 17.4 | 9.9 | | |

cases for category 1 (Donabedian) was a simple majority of 3/5 agreement not achieved. For category 2 (target), the lack of a simple majority rose to 10% (Table 4). Despite considerable effort being invested in producing clear definitions of the subcategories, the kappa results suggest that some residual ambiguity remains when interpreting the standards.

Differences were observed in the kappa agreements between documents. Some appeared to be more amenable to interpretation than others, as evidenced by a higher kappa. There may be lessons here on how to write quality statements whose purpose and intent are clear.

Overall the 2-step decision tree-based approach led to the collation of a large number of standards into 12 lists of average size, 40 which are manageable when looking for duplicates or identifying gaps in current documents. Inter-rater agreement, assessed using kappa statistics, was good considering the complexity of the task.

## Conclusions

The approach described has served to efficiently, and with relatively high reliability, separate 454 quality standards from 8 national and international documents. This separation into 3 × 4 = 12 categories, with manageable numbers of standards in each, facilitates the continued development of quality documents for use in radiation medicine. Complete tabulated results, which will be useful to others engaged in quality assessment in radiation medicine, are available from the electronic archive (available as supplementary material online only at www.practicalradonc.org). Given that there is no clear "best practice" for revising guidelines and standards, the described methodology may be useful for other groups interested in a formal consensus process to inform document redevelopment. This process has reaffirmed the robustness of Donabedian's model as a useful guide for managing quality indicators.

The collation resulting from this study is being used as one input in the refinement of the Canadian document *Quality Assurance Guidance for Canadian Radiation Treatment Programs*.

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